ADSP 31006 IP02
Time Series Analysis and Forecasting
FINAL PROJECT

U.S. Natural Gas Consumption Forecasting



Group 7

- Rui Gong:
  - Experimental results and Analysis
  - Benchmark: Seasonal Naive
- Wade Chen:
  - Regression: Linear Trend with Seasonal Dummies
- Leo Liu
  - ARIMA Modeling
- Zihao Huang:
  - Exponential Smoothing: Holt-Winters Seasonal Method
  - Summary

Project frame

- Dataset: U.S. Natural Gas Total Consumption
- Frequency: Monthly
- Study window: 2006-01 to 2025-12
- Train / test: 216 / 24 observations

Introduction

Business Problem

U.S. natural gas consumption forecasting is an important business problem because accurate demand forecasts can support energy planning, infrastructure management, storage and supply decisions, and broader policy analysis.

Since natural gas demand is highly seasonal and can change over time, the key question of this project is whether classical time-series methods can produce reliable short- and medium-term forecasts for monthly U.S. total natural gas consumption.

Data Source: https://www.eia.gov/dnav/ng/hist/n9140us2m.htm

Target Variable: U.S. Natural Gas Total Consumption (Million Cubic Feet)

Train / Test split:
Training set: Jan 2006 – Dec 2023 (216 months)
Testing set: Jan 2024 – Dec 2025 (24 months)

Agenda

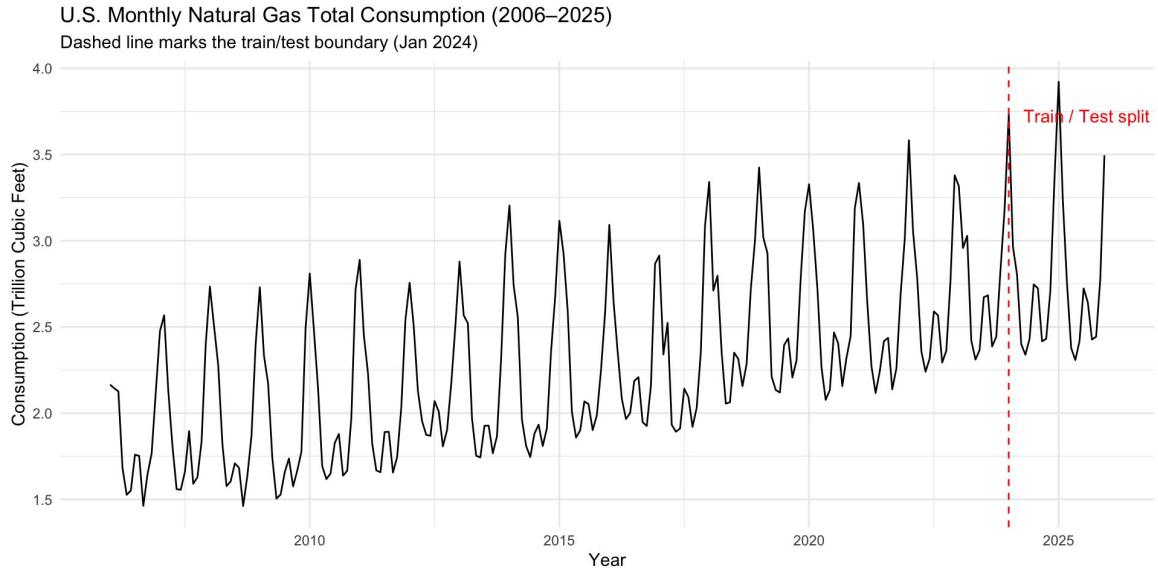
1. Experimental results and Analysis
2. Benchmark Model: Seasonal Naïve
3. Regression Model: Linear Trend with Seasonal Dummies
4. ARIMA
5. Exponential Smoothing: Holt-Winters Seasonal Method
6. Final Model Comparison and Conclusion

Part 1 — Experimental results and Analysis



Data preparation, descriptive structure, transformation choice, and dependence diagnostics

1.1 Time Series Plot



Interpretation

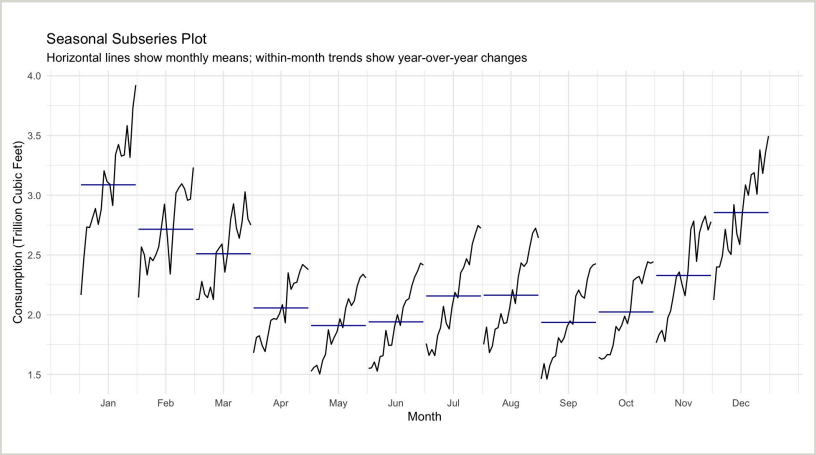
- A clear upward level shift is visible across the 20-year span.
- Winter peaks and shoulder-season troughs indicate strong annual seasonality.
- Seasonal amplitude grows with the level, which hints at multiplicative variance.
- Two notable disruptions are highlighted in the notebook narrative: the 2014 Polar Vortex and the 2020 COVID-era dip.

Main takeaway

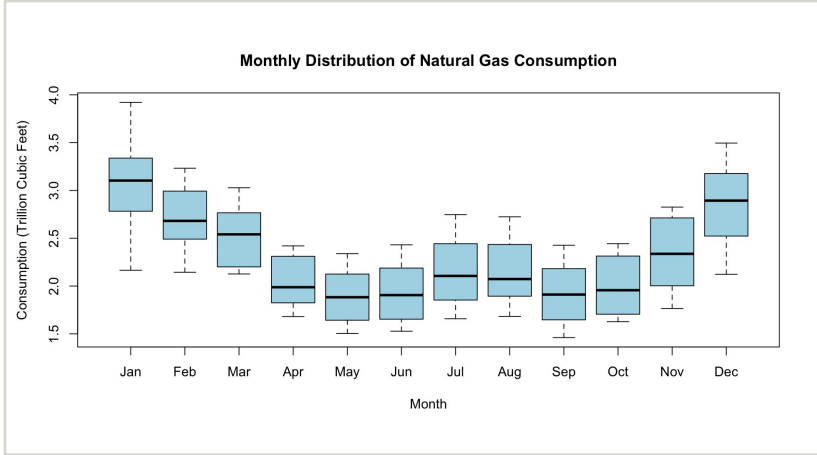
The series is seasonal, trending, and likely heteroskedastic — so descriptive transformation diagnostics are necessary before model comparison.

1.2 Seasonal Structure

Seasonal subseries

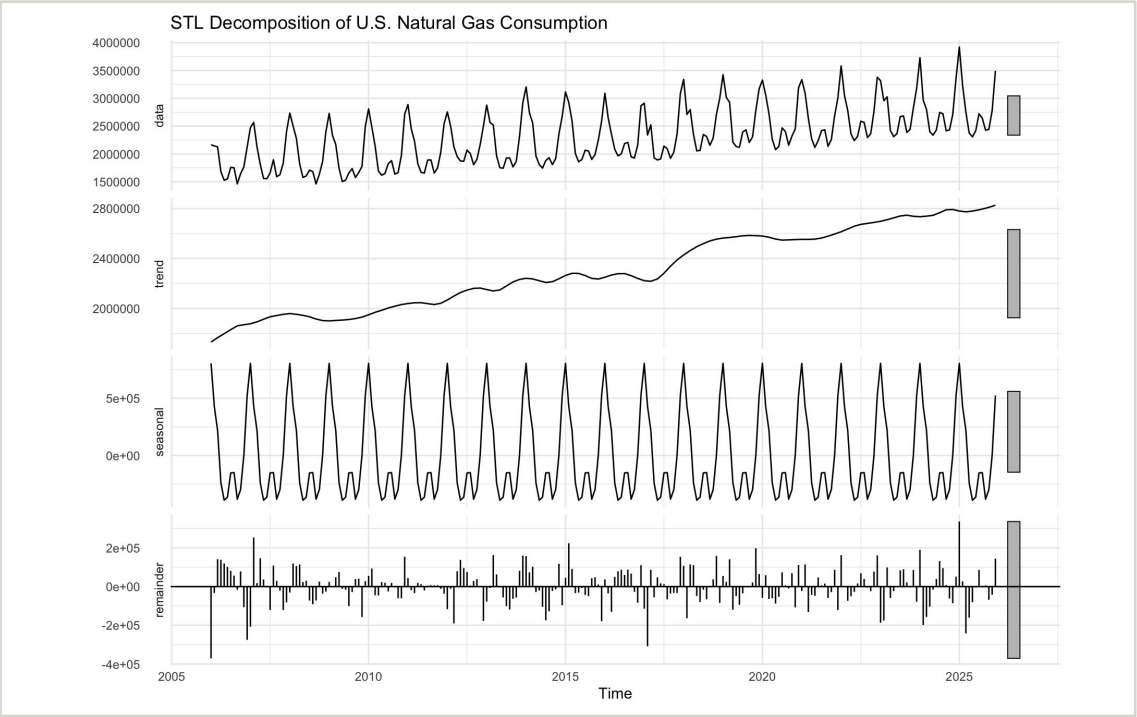


Monthly box plot



- January is the highest-consumption month; December and February are the next most elevated winter months.
- September and October behave as the lowest-demand shoulder-season months.
- Within-month trajectories trend upward over time, and winter months also show the widest spread, reinforcing the variance-growth story.

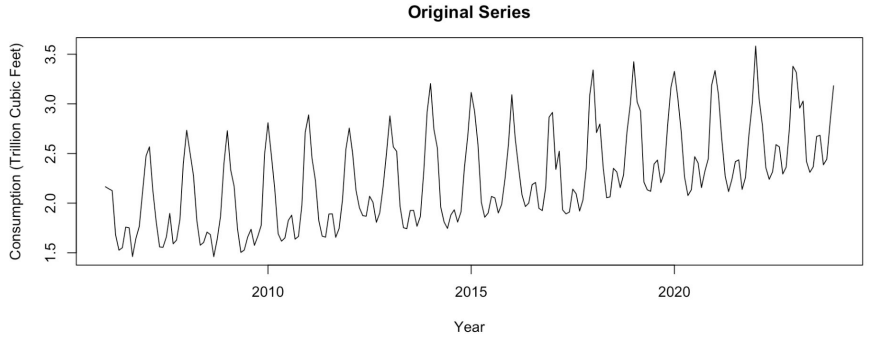
1.3 STL Decomposition

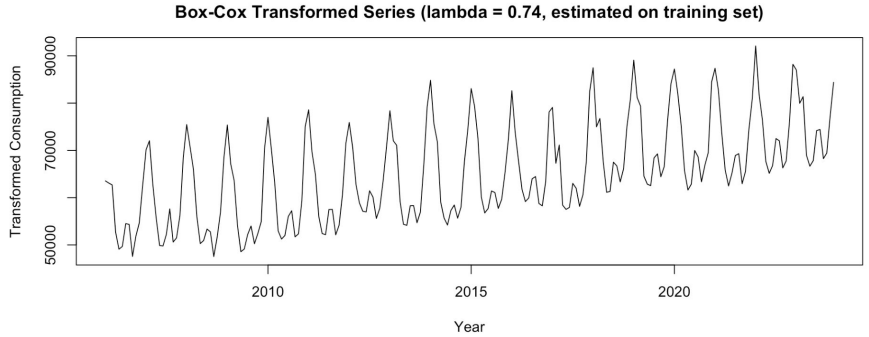


Interpretation

- Trend rises from roughly 1.8 to 2.7 trillion cubic feet per month over the study period.
- The seasonal component is persistent and large, with annual amplitude around ±500–700 billion cubic feet.
- The remainder is mostly noisy, though large shocks remain visible around early 2014 and 2020.

1.4 Box–Cox Transformation Analysis





Transformation rule

$$y_t^{\lambda} = (y_t^{\lambda} - 1)/\lambda, \quad \lambda \neq 0$$

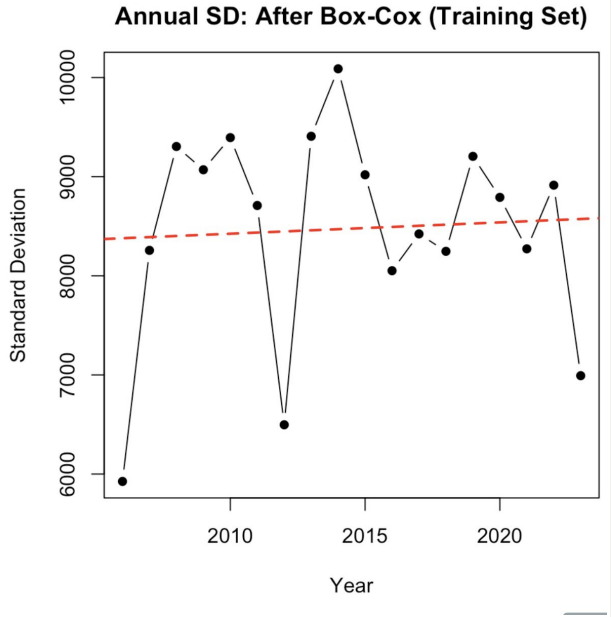
and $\log(y_t)$ when λ = 0

- Estimated λ = 0.7372, so the notebook recommends a Box–Cox transform rather than no transform or a pure log transform.
- The transformed plot compresses larger peaks and reduces the visible growth in seasonal amplitude.
- Team handoff: all downstream models should use the same λ for consistency.

Decision
Apply Box–Cox with λ = 0.7372 in the shared modeling pipeline.

1.5 Variance Stabilization Check





Interpretation

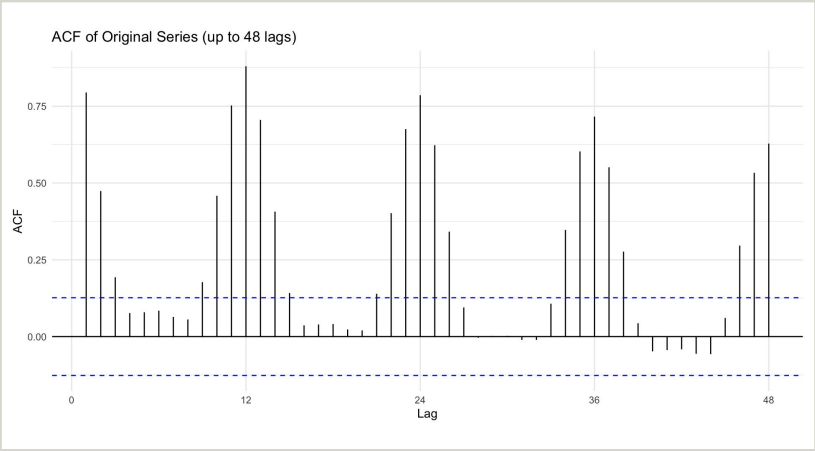
- Before transformation, annual standard deviation varies substantially over time.
- After Box–Cox, the year-by-year spread is visibly flatter, which is consistent with better variance stability.
- This visual check supports keeping the transformed series for later models.

Bottom line

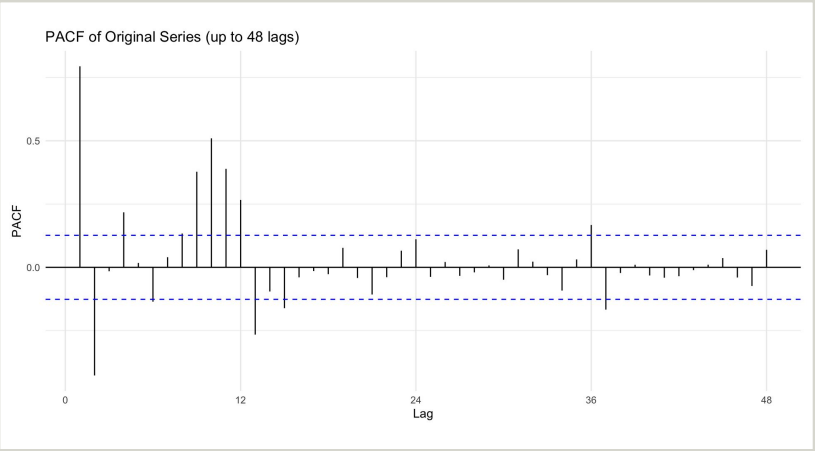
The transform is not cosmetic; it directly targets heteroskedasticity in the observed seasonal pattern.

1.6 ACF and PACF of the Original Series

ACF

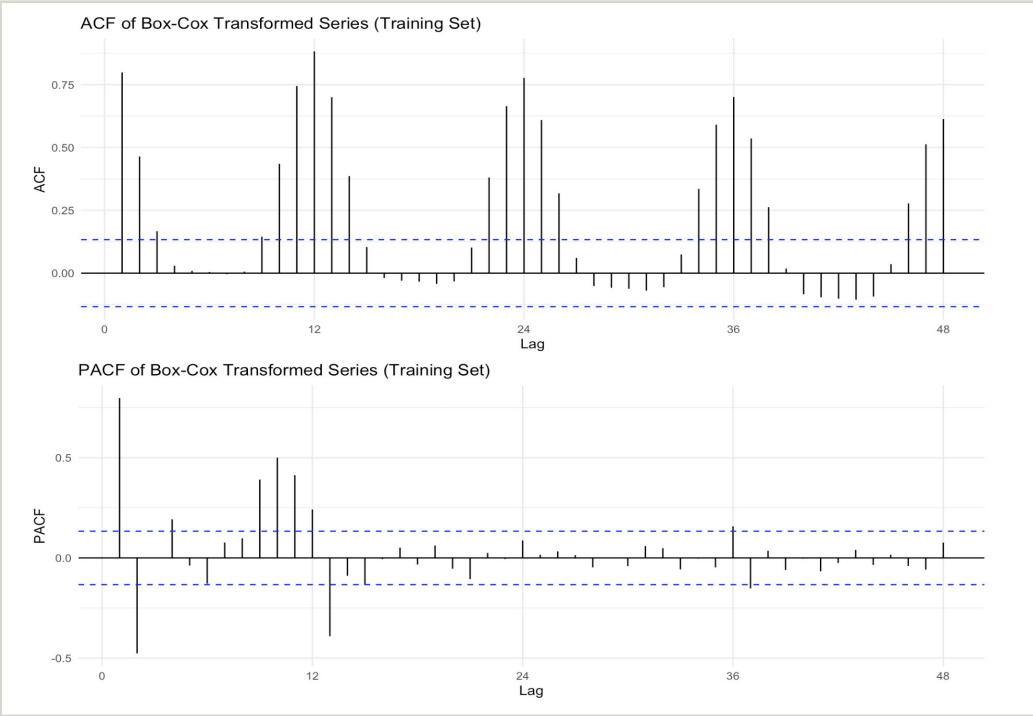


PACF



- The ACF decays very slowly and has pronounced spikes at lags 12, 24, 36, and 48 — classic evidence of trend plus annual seasonality.
- The PACF shows a dominant lag-1 spike and a notable lag-12 spike, which suggests both short-run and seasonal dependence.
- These diagnostics strongly motivate regular plus seasonal differencing for later ARIMA modeling.

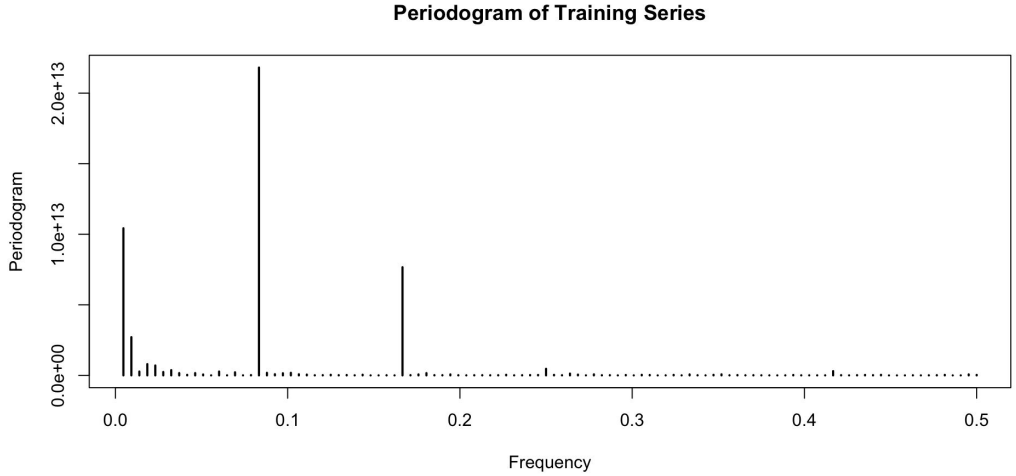
1.7 ACF / PACF of the Transformed Series



Interpretation

- The overall ACF/PACF patterns remain similar to the original series — slow decay in ACF with seasonal peaks, and dominant PACF spikes at lags 1 and 12. This is expected because the Box-Cox transformation stabilizes **variance**, not the mean structure or autocorrelation.
- The temporal dependency structure (trend + seasonality) persists after transformation, confirming that variance stabilization alone is not sufficient to achieve stationarity — differencing would still be required for models that assume stationary input.

1.8 Spectral Analysis



Top frequencies from the notebook

Frequency	Spectrum	Period (months)
0.0833	2.18E13	12
0.00463	1.04E13	216
0.1667	7.66E12	6
0.00926	2.71E12	108
0.0185	7.99E11	54

- The dominant spectral peak occurs near f ≈ 1/12, confirming a 12-month annual cycle.
- A second visible peak near f ≈ 1/6 is a semi-annual harmonic, which often appears when seasonal shape is not purely sinusoidal.

Part 2 — Benchmark Model: Seasonal Naïve



Benchmark model for comparing later ARIMA, ETS, regression, and other model classes

2.1 Definition and Fit

Model definition

$\hat{y}_{t+h} = y_{t+h-m}$, $m = 12$

- Forecasts reuse the most recent observation from the same month one year earlier.
- For monthly data, this is the simplest seasonal benchmark that any stronger model should beat.

Training-set fit summary

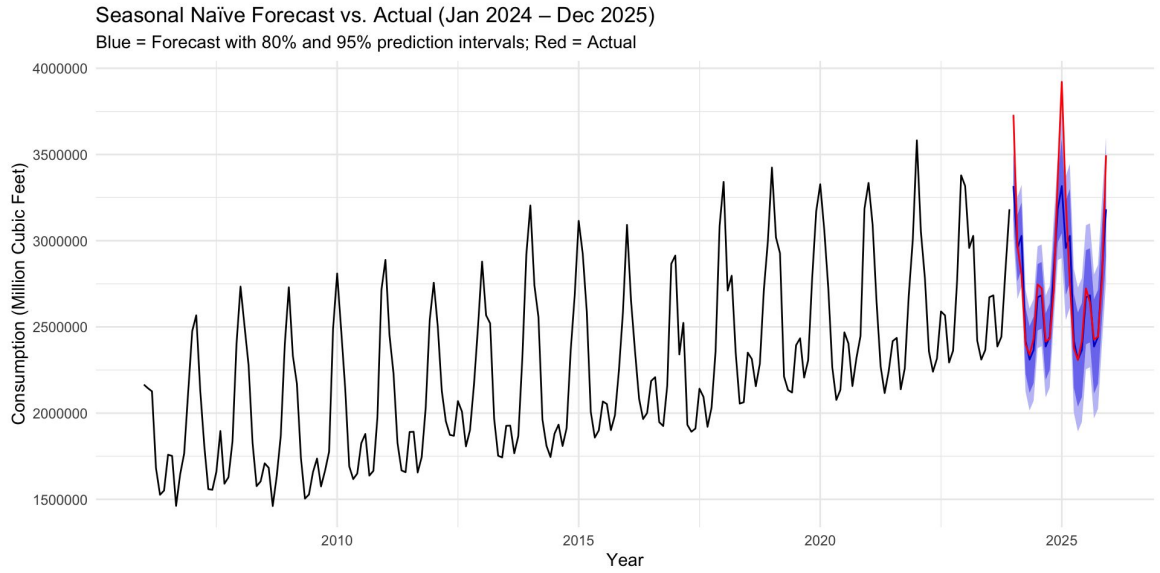
Residual sd	150,598.6
RMSE	150,598.6
MAE	117,395.4
MAPE	5.04%
MASE	1.00
ACF1	0.3325

Forecast setup

- Fit on training data only.
- Forecast horizon h = 24 months.
- Benchmark against Jan 2024–Dec 2025 test set.

This is the simplest reasonable benchmark for seasonal time series and serves as the baseline for comparing more sophisticated models.

2.2 Forecast Plot

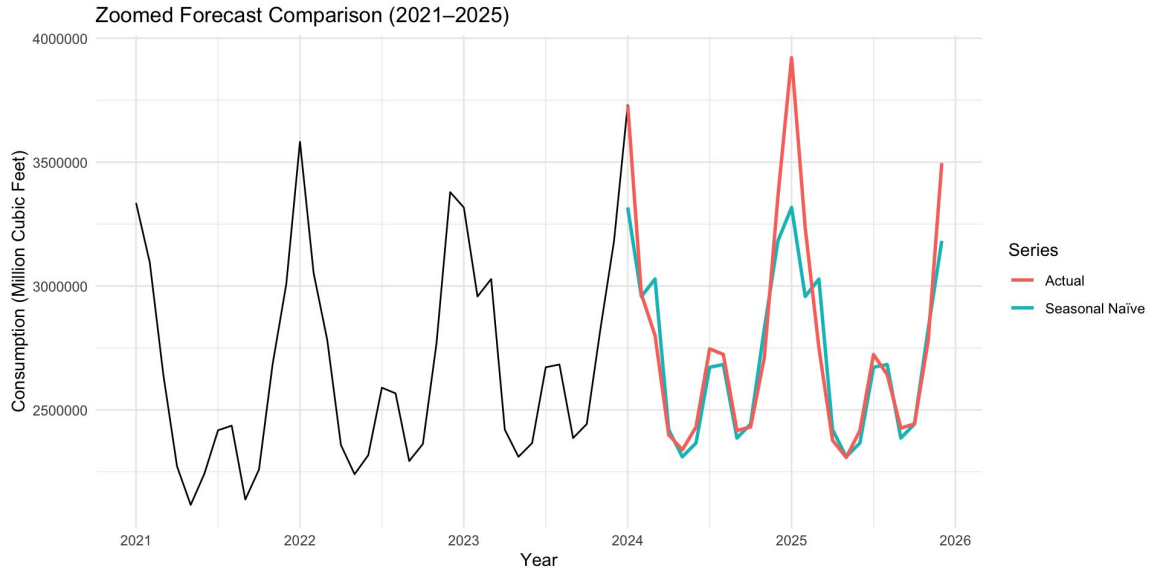


Interpretation

- The forecast path replicates the 2023 seasonal profile into 2024 and 2025.
- Prediction intervals capture some uncertainty, but misses remain where the actual series departs from repeated seasonality.
- This is exactly what we want from a baseline: it is interpretable, but clearly limited.

This figure is used as the visual baseline that later models need to surpass in both fit and adaptability.

2.3 Zoomed Forecast Comparison



Close-up reading

- The benchmark tracks the broad seasonal shape but cannot adapt to year-to-year magnitude changes.
- Discrepancies are clearest around high peaks and sharp rebounds, where copied seasonality is too rigid.
- This zoomed view makes the benchmark limitation visible without relying only on aggregate metrics.

Practical conclusion

The model captures “when” demand rises and falls, but not always “how much.”

2.4 Forecast Accuracy

Test RMSE

194,709

Original units

Test MAE

123,366

Original units

Test MAPE

3.90%

Percentage error

Test MASE

1.0509

Relative to seasonal-naïve scale

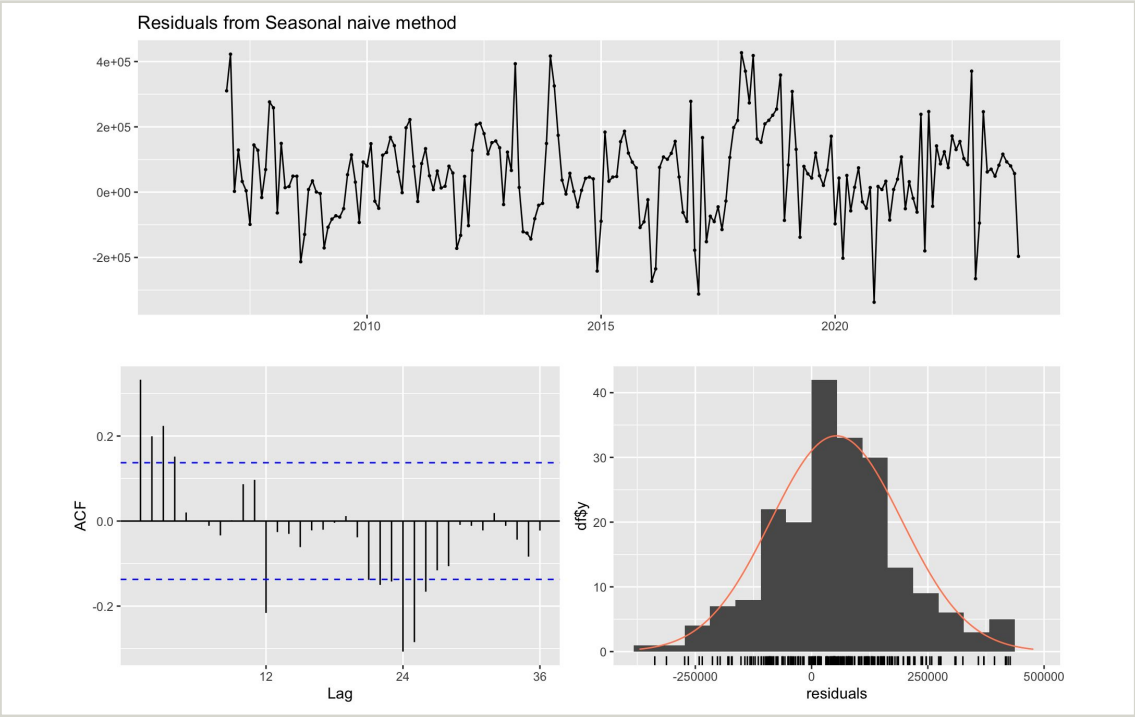
Training vs. test summary

Set	RMSE	MAE	MAPE	MASE	Theil's U
Training	150,599	117,395	5.04%	1.00	—
Test	194,709	123,366	3.90%	1.05	0.44

Interpretation

- **Good baseline:** average error is about **123 Bcf/month**.
- **Strong seasonality: MAPE = 3.90%** shows seasonal patterns explain much of the variation.
- **Can be improved: MASE = 1.05** suggests slight weakness from missing the upward trend.

2.5 Residual Diagnostics——Comprehensive Residual Check



Ljung–Box test

Q* = 98.754 | df = 24
p-value = 4.896e-11

Interpretation

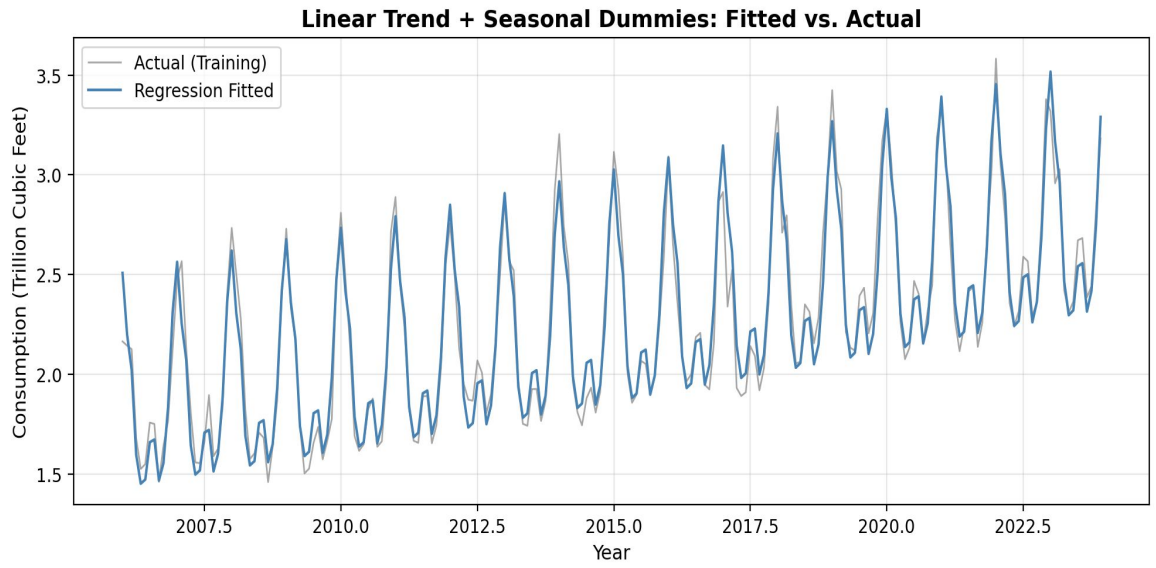
- Residual autocorrelation remains visible, so the benchmark does not explain all serial structure.
- The very small Ljung–Box p-value rejects white-noise residuals.
- That is expected for a deliberately simple baseline and motivates richer models.

Part 3 — Regression Model: Linear Trend with Seasonal Dummies



Data preparation, descriptive structure, transformation choice, and dependence diagnostics

3.1 Model Specification and Fit



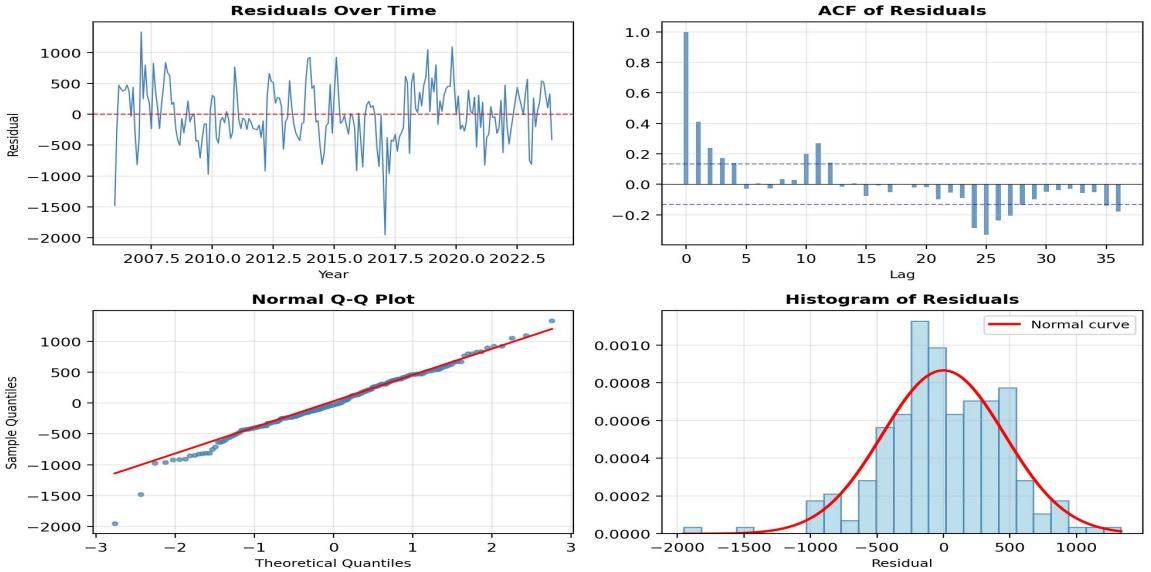
Interpretation

- Model: tslm(y ~ trend + season) on Box-Cox transformed data (lambda = 0.63).
- 13 parameters: intercept + linear trend + 11 monthly dummies (January = reference).
- R-squared = 0.95: trend and seasonal dummies explain most variance in transformed series.

Main takeaway

The fitted values track the seasonal shape well. Year-to-year deviations (e.g. 2014 Polar Vortex, 2020 COVID dip) remain in the residuals.

3.2 Residual Diagnostics



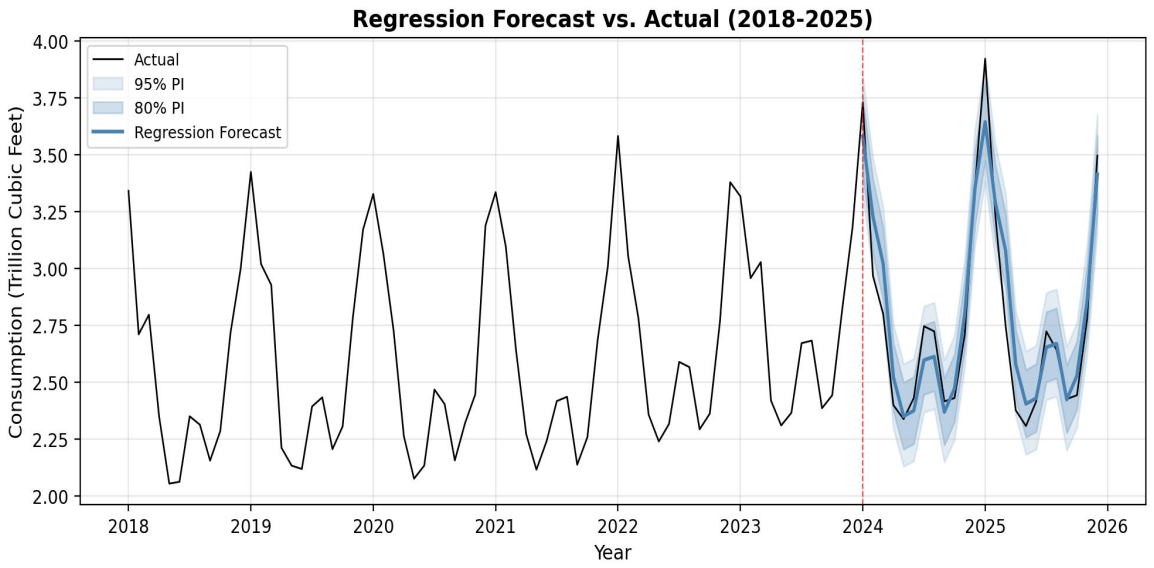
Interpretation

- ACF shows significant spikes at lags 1, 12, and 24 — residual autocorrelation remains after regression.
- Durbin-Watson test rejects H0 ($p < 0.05$): positive lag-1 autocorrelation confirmed.
- Breusch-Godfrey test (order 12) also rejects H0: seasonal autocorrelation persists in residuals.

Main takeaway

Forecasts are unbiased but prediction intervals are wider than necessary. ARIMA and ETS can exploit this leftover structure.

3.3 Forecast and Accuracy



Interpretation

- RMSE: 139,749 vs. benchmark 194,709 (28% improvement).
- MAPE: 3.83% vs. benchmark 3.90%.
- MASE: 0.92 (< 1.0) — regression outperforms seasonal naive on scaled error.
- Wide prediction intervals reflect unmodeled autocorrelation in residuals.

Main takeaway

Regression improves on the benchmark by capturing trend. Autocorrelated residuals motivate ARIMA (Part 4) and ETS (Part 5).

Part 4 — ARIMA Model



Stationarity transformation, model identification, selection, and diagnostic checking

4.1 ARIMA Model

SARIMA Model Structure

$$\Phi(B^{12}) \cdot \phi(B) \cdot (1-B)^{d} \cdot (1-B^{12})^{D} y_t = \Theta(B^{12}) \cdot \theta(B) \varepsilon_t$$

where B is the backshift operator and the seasonal period is 12

AR(p)	I(d, D)	MA(q)	Seasonal $(P,D,Q)_{12}$
p = 0	d = 0 D = 1	q = 1	P = 0, Q = 1
Uses past values of y_t Captures persistence	Regular differencing: d Seasonal differencing: D at lag 12 Makes the series stationary	Uses past forecast errors Captures short-term correction	Seasonal terms at lag 12 Captures yearly pattern

Selected model

$$\text{SARIMA}(0, 0, 1)(0, 1, 1)_{12}$$

Box-Cox transformation: $\lambda = 0.7372$

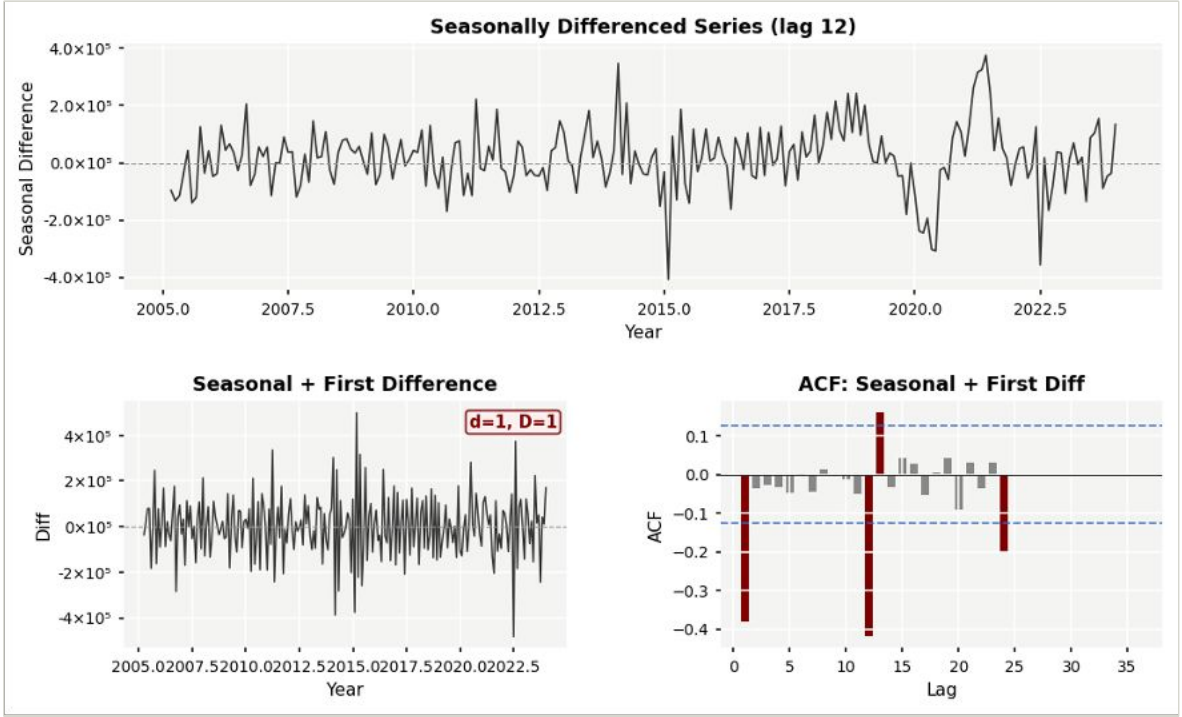
Interpretation

- auto.arima() selects ARIMA(0,0,1)(0,1,1)[12] with Box-Cox λ=0.7372
- MA(1)=0.4239: corrects 42% of prior-month forecast error
- SMA(1)= -0.3021: dampens same-month-last-year error by 30%
- No AR terms: series after differencing is MA-driven

Main takeaway

Parsimonious SARIMA with one MA + one seasonal MA captures both short-run dynamics and the 12-month seasonal cycle.

4.2 Differencing & Stationarity



Interpretation

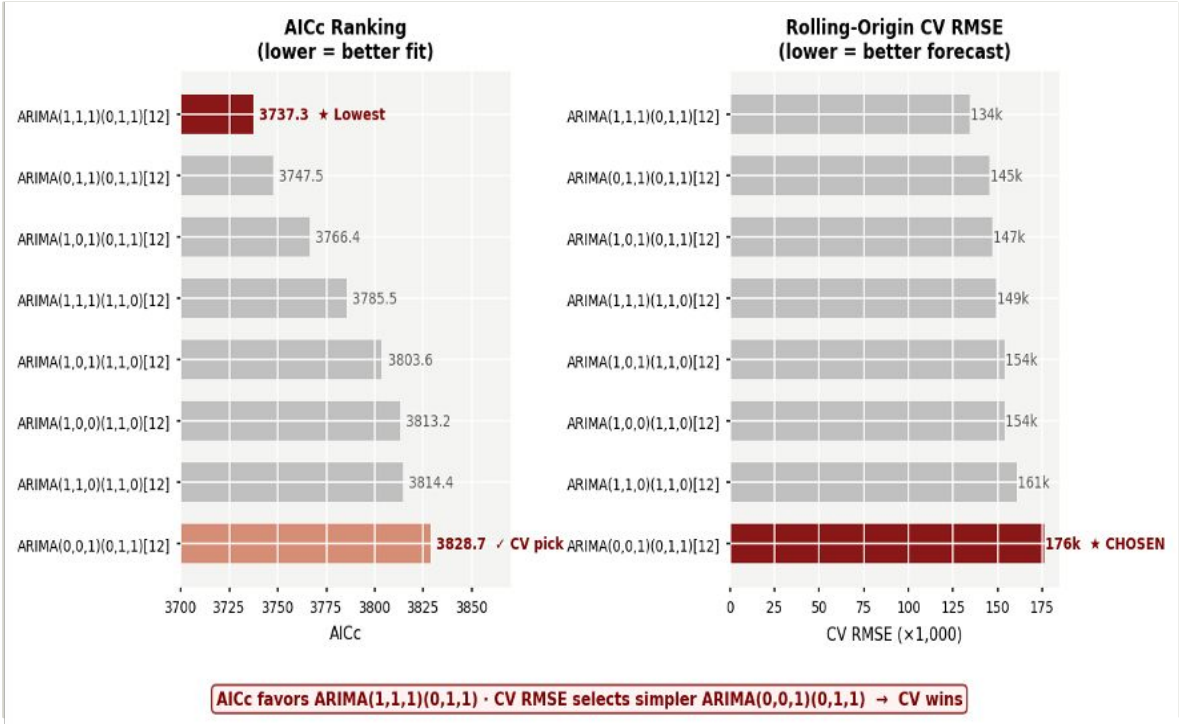
- ADF: Dickey-Fuller=-11.656, p=0.01 → reject unit root
- KPSS Trend=0.015, p=0.10 → fail to reject stationarity

- ndiffs()=1: one regular difference needed
- nsdiffs()=1: one seasonal difference (lag 12) needed
- After D=1, d=1: series fluctuates around zero

Main takeaway

Both ADF and KPSS agree: Box-Cox series needs d=1, D=1 to achieve full stationarity for ARIMA fitting.

4.3 Model Selection



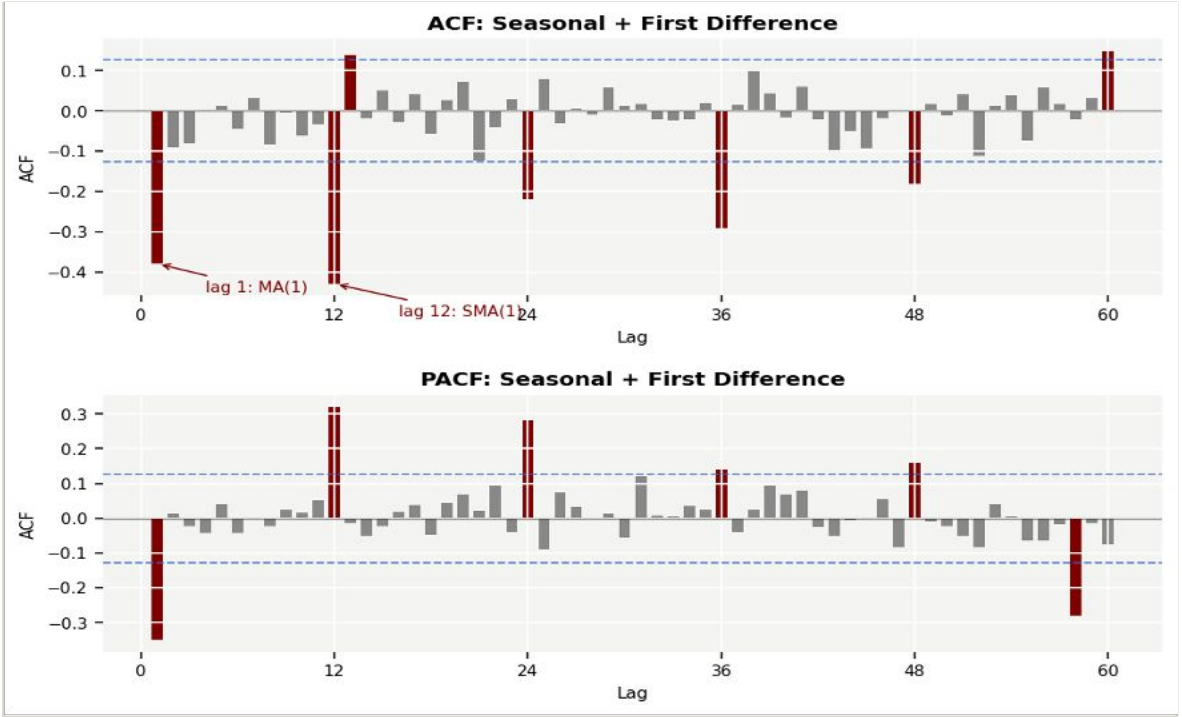
Interpretation

- 8 candidate SARIMA models evaluated
- Lowest AICc: ARIMA(1,1,1)(0,1,1)[12] = 3737.3
- Best CV RMSE: ARIMA(0,0,1)(0,1,1)[12] = 149,301
- AICc and CV diverge → CV wins for forecasting
- auto.arima() confirms ARIMA(0,0,1)(0,1,1)[12]

Main takeaway

Cross-validation, not AICc alone, drives model choice. The parsimonious SARIMA(0,0,1)(0,1,1)[12] generalises better out-of-sample.

4.4 ACF/PACF Pattern Analysis



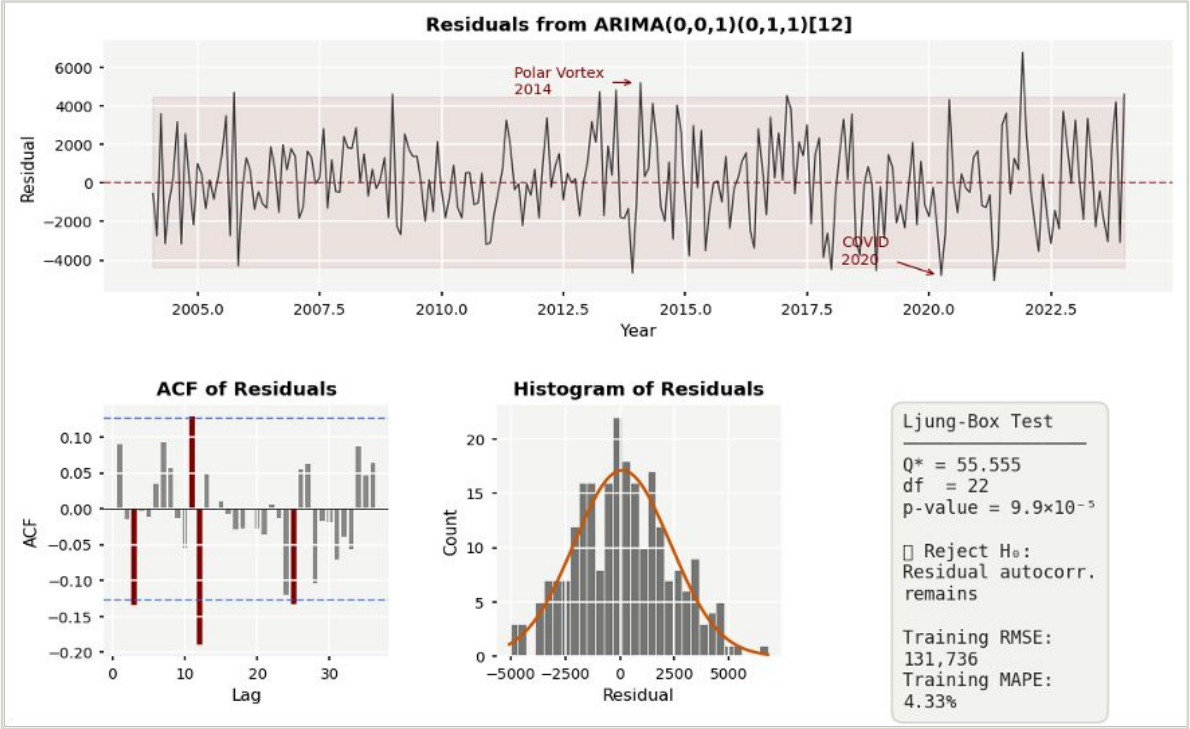
Interpretation

- ACF lag 1 cut-off → MA(1) term appropriate
- ACF lag 12 significant → seasonal MA(1) needed
- PACF decays slowly → no AR term required
- Both ACF and PACF suggest ARIMA(0,d,1)(0,D,1)[12]
- Pattern consistent with all 8 candidate models

Main takeaway

ACF/PACF patterns clearly suggest one MA term and one seasonal MA term — matching the final selected model.

4.5 Residual Diagnostics



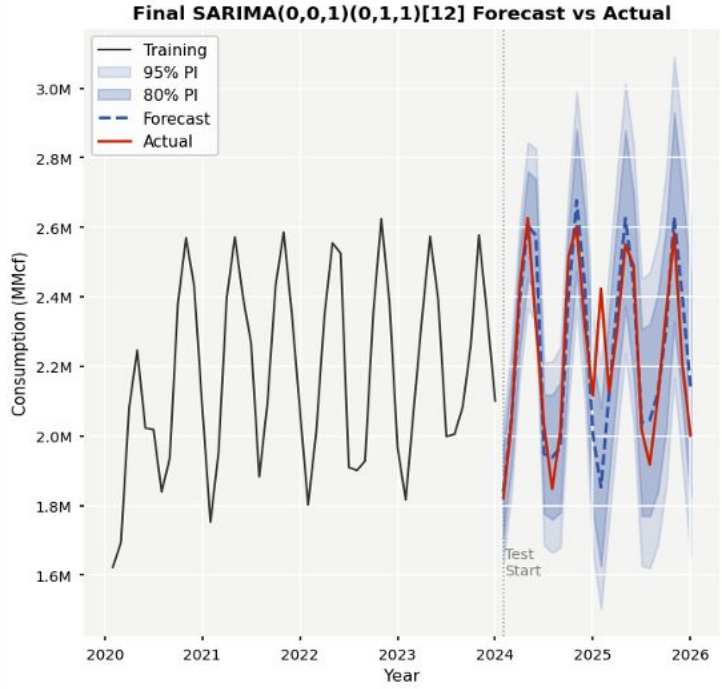
Interpretation

- Residuals fluctuate around zero — model unbiased
- Histogram approximately normal, slight heavy tails
- ACF: most lags within 95% bands
- Ljung-Box $Q^*=55.6$, df=22, $p=9.9\times10^{-5}$
- Residual autocorrelation remains — known limitation

Main takeaway

Diagnostics are reasonable: residuals are **near-normal** and **unbiased**. Ljung-Box signals some remaining structure, a known limitation of parsimonious SARIMA.

4.6 Residual Diagnostics



Forecast Accuracy

Metric	Train	Test
RMSE	131,736	182,735
MAE	102,864	121,987
MAPE	4.33%	3.92%
MASE	0.876	1.039
Theil U	—	0.414

Theil U = 0.414 < 1

Model beats naive benchmark on all windows

Interpretation

- Test MAPE = 3.92%: strong out-of-sample performance
- Test RMSE = 182,735 Test MAE = 121,987
- Theil U = 0.414 < 1: beats naive on all windows
- Jan 2025 peak underestimated (extreme outlier)
- Training MAPE = 4.33%: no overfitting

Main takeaway

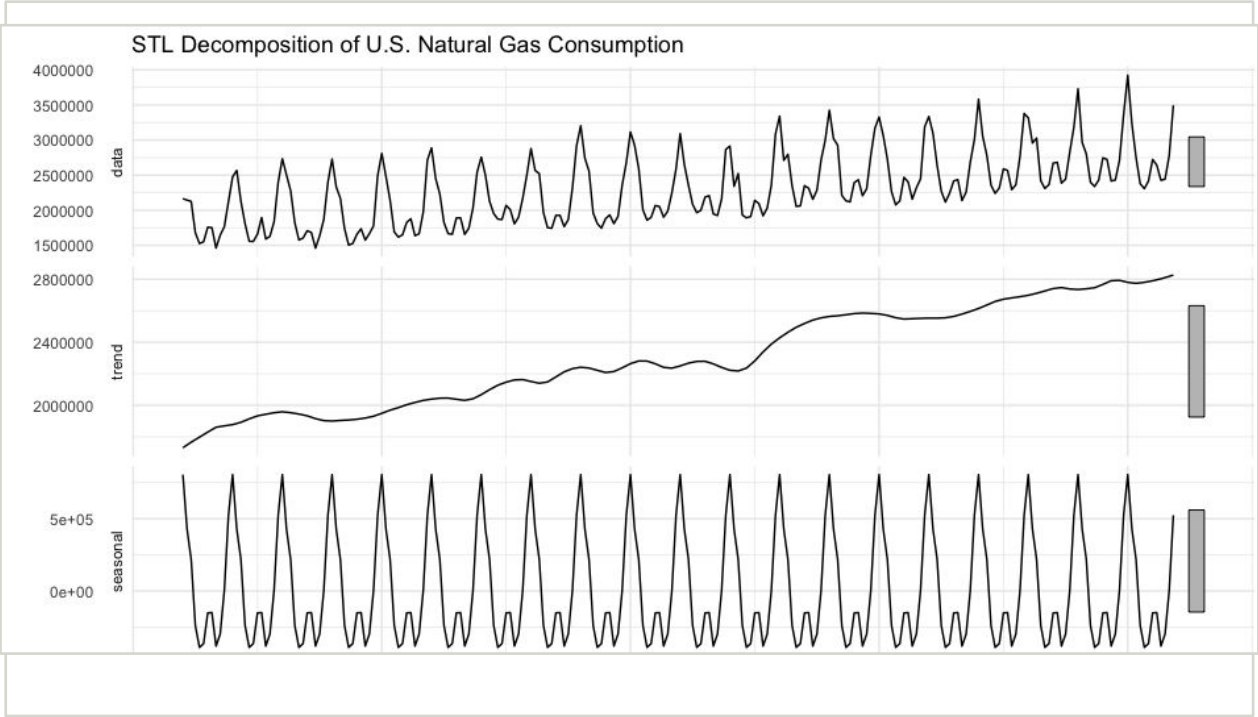
Test MAPE = 3.92%, Theil U = 0.414 — beats naive benchmark on all windows.

Part 5 — Exponential Smoothing: Holt-Winters Seasonal Method



Data preparation, descriptive structure, transformation choice, and dependence diagnostics

5.1 Model Specification and Fit



Intuition

Holt-Winters uses recursive smoothing to update level, trend, and seasonality, and is well suited for series with evolving trend and seasonal pattern.

Key Components

- level: current baseline
- trend: growth over time
- seasonality: repeating 12-month pattern

- forecast = combination of level + trend + seasonality

5.2 Model Selection

- To identify the most suitable Holt-Winters specification, we compared several candidate models with different seasonal and trend assumptions.
- We considered both additive and multiplicative seasonality, since the series shows strong yearly seasonality and the seasonal amplitude may vary with the level of the series.
- Therefore, we have the following candidates:
  - Linear trend with additive seasonality and non-damped
  - Linear trend with additive seasonality and damped
  - Linear trend with additive seasonality, non-damped, and Box-Cox transformation
  - Linear trend with additive seasonality, damped, and Box-Cox transformation
  - Linear trend with multiplicative seasonality and non-damped
  - Linear trend with multiplicative seasonality and damped

5.3 Model Selection

Holt-Winters Model Comparison

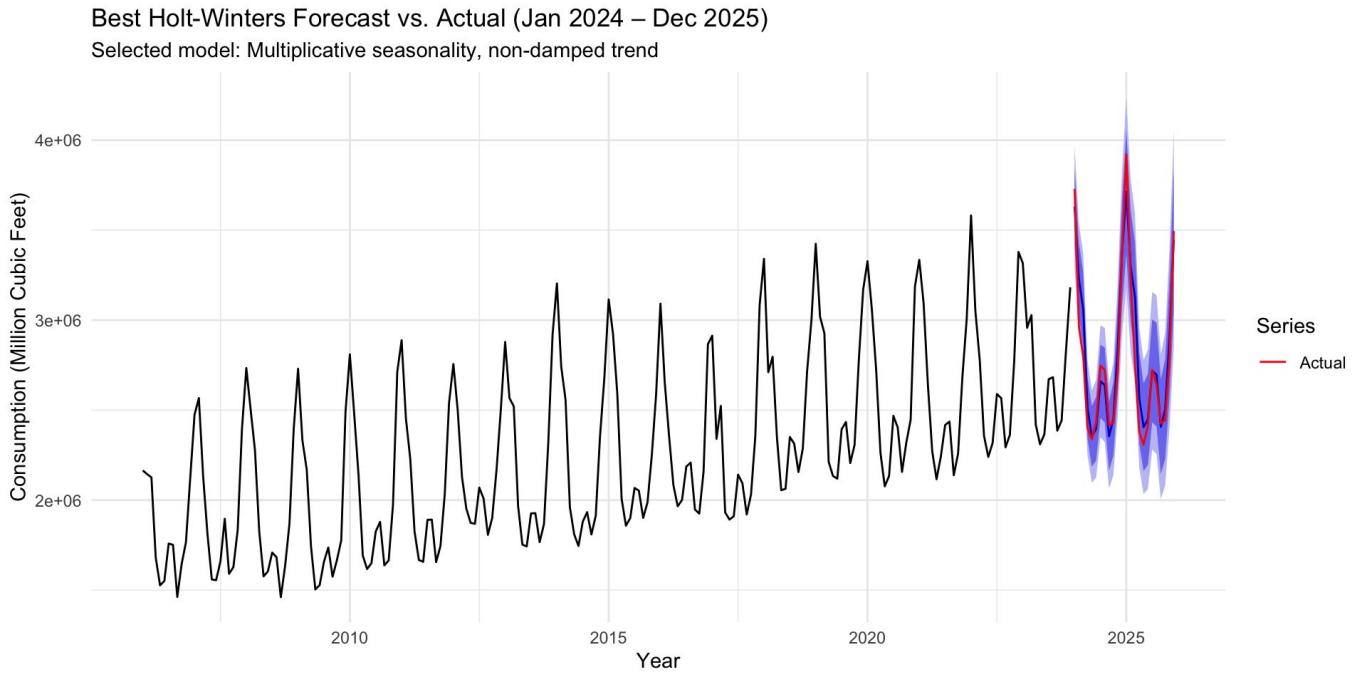
	AICc	RMSE	MAE	MAPE (%)
HW_Add	6199.108	154642.1	125423.1	4.523928
HW_Add_Damped	6199.508	149165.4	116613.8	3.983908
HW_Add_BC	4525.092	152400.6	121877.1	4.409761
HW_Add_Damped_BC	4526.054	149036.3	116159.6	4.004398
HW_Mul	**6173.591**	**138222.9**	**102544.9**	**3.679486**
HW_Mul_Damped	6175.927	152729.9	128645.3	4.654160

Note:
Blue indicates the best model based on the lowest test-set RMSE. AICc is reported as a supplementary in-sample metric.

Interpretation

- HW_Mul achieved the lowest test-set RMSE of 138,222.9, lowest MAE of 102,544.9, and the lowest MAPE of 3.68%, making it the best Holt-Winters specification.
- Multiplicative seasonality performed better than additive seasonality, suggesting that seasonal fluctuations increase with the level of the series.
- Adding trend damping did not improve performance, since HW_Mul_Damped was notably worse than HW_Mul across all test-set error metrics.

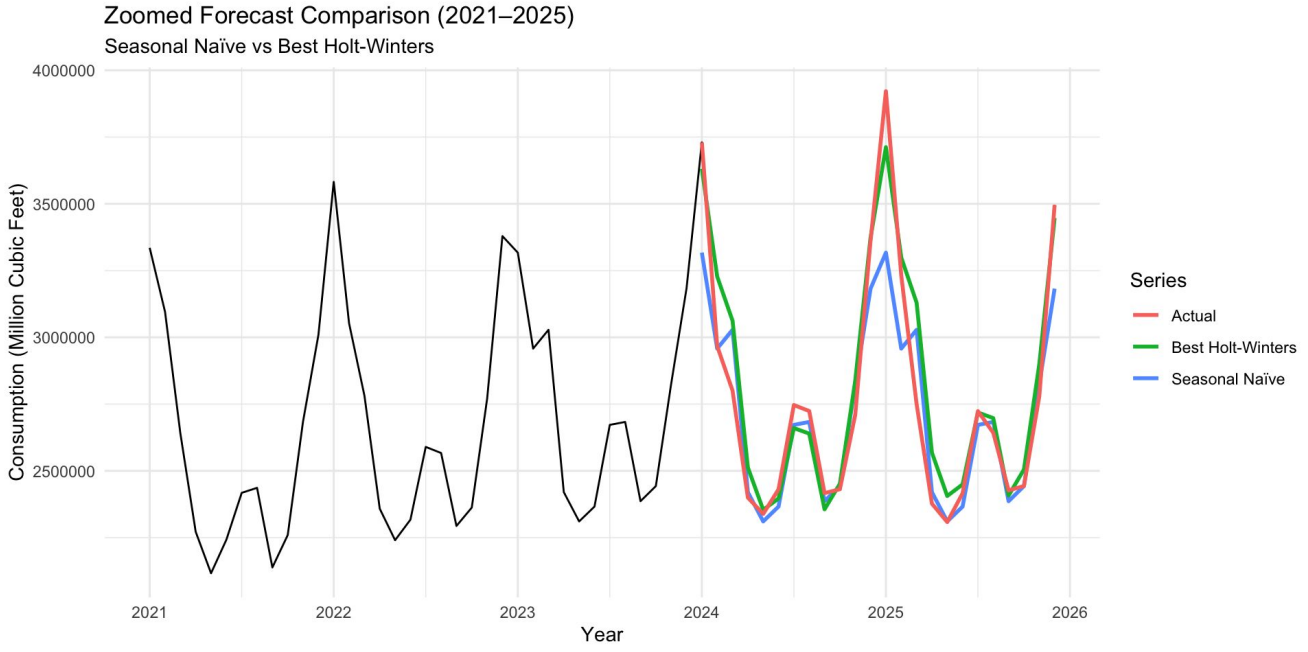
5.4 Forecast



Interpretation

- The selected Holt-Winters model tracks the actual series closely over the 2024–2025 test period.
- It captures the main seasonal pattern well, including recurring peaks and troughs.
- Prediction intervals widen over time, reflecting increasing forecast uncertainty.

5.4 Forecast



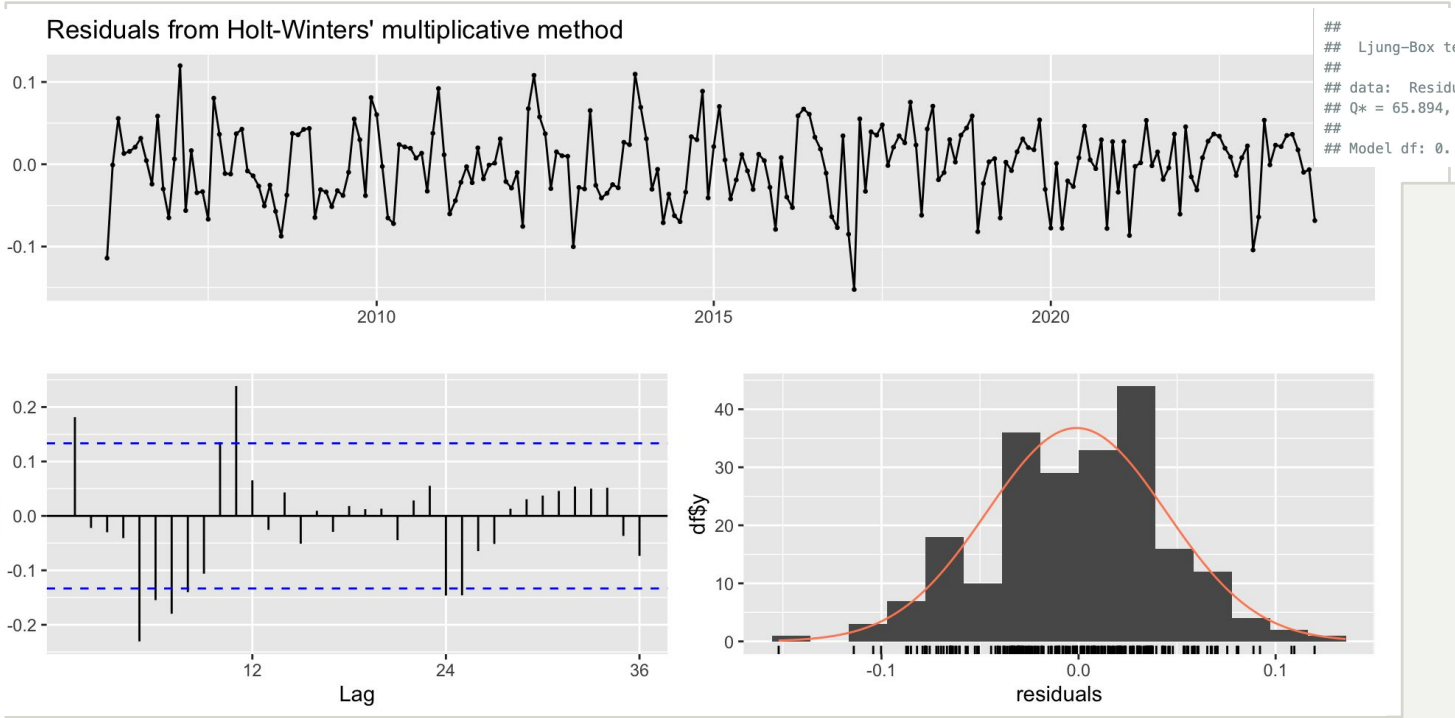
Interpretation

- Compared with Seasonal Naïve, Holt-Winters stays closer to the actual series in most months.
- This suggests that modeling both evolving level and multiplicative seasonality improves forecast accuracy.

Main takeaway

The selected Holt-Winters model captures the series' seasonal pattern and changing level well, producing forecasts that stay close to the actual values. It also outperforms the Seasonal Naïve benchmark, showing the value of modeling trend and multiplicative seasonality together.

5.5 Residual Diagnosis




```
##
##  Ljung-Box test
##
## data:  Residuals from Holt-Winters' multiplicative method
## Q* = 65.894, df = 24, p-value = 8.993e-06
##
## Model df: 0.   Total lags used: 24
```

-The residuals fluctuate around zero with no obvious long-term trend
-the ACF plot shows several spikes outside the confidence bounds, especially at seasonal lags
-The Ljung–Box test strongly rejects the null of no autocorrelation (p-value < 0.001), so the residuals are not white noise.
-The Holt-Winters model captures the main pattern well, but the residual diagnostics show that some autocorrelation remains, so the fit is good but not perfect.

Part 6 — Final Model Comparison and Conclusion



Data preparation, descriptive structure, transformation choice, and dependence diagnostics

6.1 Models Comparison

Test-Set Accuracy: Final Models vs. Seasonal Naïve Benchmark

Model	RMSE	MAE	MAPE (%)	MASE
Seasonal Naïve (Benchmark)	194708.5	123366.4	3.90	1.0509
Regression (Trend + Season)	140262.7	109256.9	3.85	0.9307
ARIMA (Selected Final Model)	182735.1	121987.1	3.92	1.0391
Holt-Winters (Selected Final Model)	**138222.9**	**102544.9**	**3.68**	**0.8735**

Note:

Pink = benchmark. Blue = best model (lowest RMSE).

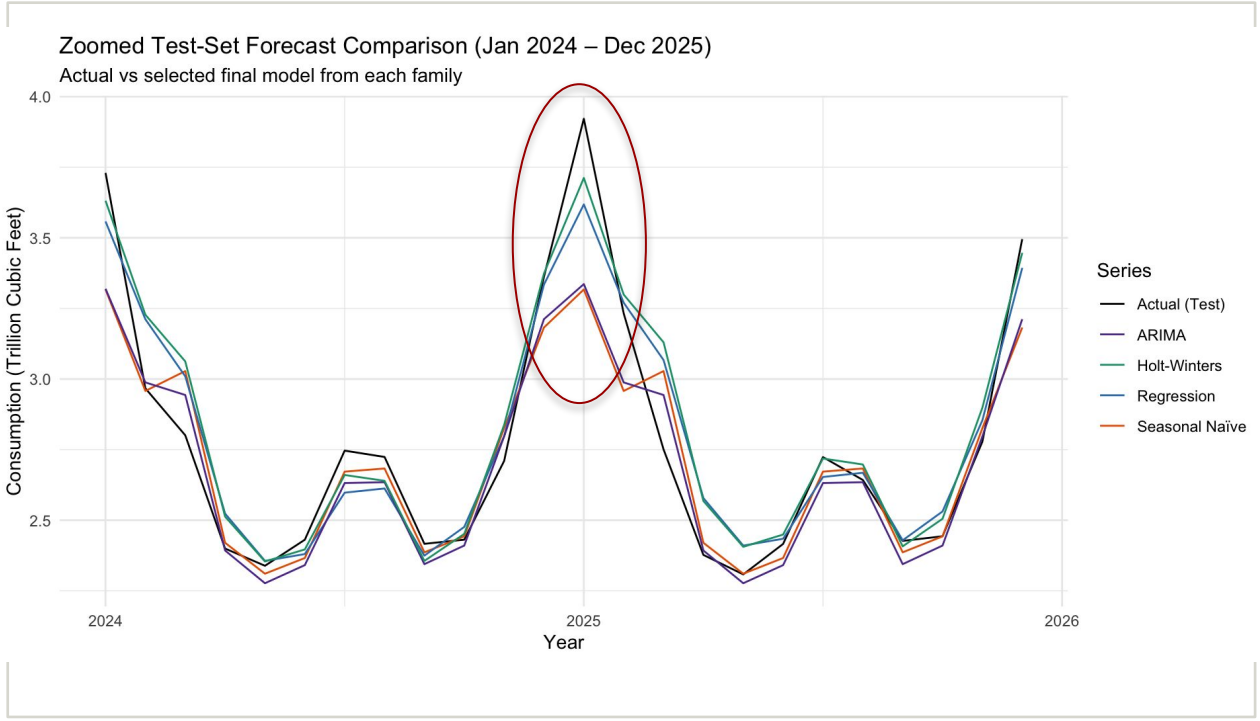
Interpretation

- Holt-Winters achieved the lowest RMSE
- It also produced the lowest MAE and MAPE

Main Takeaway

Holt-Winters was the best final model because it captured both the changing level and strong seasonality of the series. It delivered the most accurate forecasts on the test set across all reported metrics.

6.2 Zoomed Test-Set Forecast Plot



Interpretation

- All models capture the overall seasonal pattern in the test period
- Around the major peak in early 2025, Holt-Winters stays closer to the actual series than Regression
- This visual result is consistent with the test-set accuracy table, where Holt-Winters achieved the best overall forecasting performance.

6.3 Conclusion

PROS:

- Holt-Winters captured the overall trend and strong yearly seasonality well
- The selected multiplicative form handled level-dependent seasonal variation effectively
- It achieved the best test-set performance among all final models
- The model is intuitive, interpretable, and easy to implement

CONS:

- Residual diagnostics show remaining autocorrelation
- The model does not fully capture some sharp peak movements
- Prediction intervals become wide over longer horizons
- Performance may weaken if the underlying pattern changes structurally

FUTURE WORK:

- Explore seasonal ARIMA or hybrid models to capture remaining dependence
- Incorporate external variables such as weather or economic factors
- Test more flexible methods for extreme peaks and sudden changes
- Update and re-evaluate the model regularly as new data become available